

Example 3.8: Gridworld from Sutton & Barto

Complete Solution by Marginalization

1 The problem statement

This is Example 3.8: Gridworld from Sutton & Barto (2nd edition, Figure 3.5, page 60)[citation:8]. The environment is a rectangular 5×5 grid. The cells of the grid correspond to the states of the environment.

At each cell, four actions are possible: north, south, east, west. These actions deterministically cause the agent to move one cell in the respective direction on the grid.

Reward structure.

- Actions that would take the agent off the grid leave its location unchanged, but also result in a reward of -1 [citation:8].
- Other actions result in a reward of 0, except those that move the agent out of the special states A and B.
- From state A, all four actions yield a reward of $+10$ and take the agent to A' .
- From state B, all four actions yield a reward of $+5$ and take the agent to B' [citation:8].

Policy. The agent selects all four actions with equal probability in all states (equiprobable random policy).

Discount factor. $\gamma = 0.9$.

Goal. Compute the state-value function $v_\pi(s)$ for this policy[citation:8].

2 The Bellman equation for policy evaluation

The Bellman evaluation equation in Sutton & Barto notation is

$$v_\pi(s) = \sum_a \pi(a | s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')].$$

For this deterministic gridworld:

- $\pi(a | s) = \frac{1}{4}$ for all a (equiprobable random policy).
- $p(s', r | s, a) = 1$ for the unique (s', r) determined by (s, a) .

So the Bellman equation simplifies to

$$v_\pi(s) = \frac{1}{4} \sum_{a \in \{N, S, E, W\}} [r(s, a) + \gamma v_\pi(s')],$$

where s' is the state reached after action a from s (which is s itself if the action would leave the grid), and $r(s, a)$ is the reward received.

3 Solving the middle cell (3,3) as a reference

For the middle cell (3,3), none of the four actions leave the grid. All transitions go to neighboring cells with reward 0.

Using the values from the neighbors:

- Upper (2,3): $v = 8.8$
- Lower (4,3): $v = 1.5$
- Left (3,2): $v = 2.3$
- Right (3,4): $v = 1.9$

$$\begin{aligned} v_\pi(3,3) &= \frac{1}{4}[(0 + 0.9 \cdot 8.8) + (0 + 0.9 \cdot 1.5) + (0 + 0.9 \cdot 2.3) + (0 + 0.9 \cdot 1.9)] \\ &= \frac{0.9}{4}(8.8 + 1.5 + 2.3 + 1.9) = \frac{0.9 \cdot 14.5}{4} = \frac{13.05}{4} = 3.2625 \approx 3.3. \end{aligned}$$

This matches the value shown in Figure 3.5[citation:1][citation:4].

4 Solving the corner cell (1,1)

The top-left cell (1,1) is special because two of the four actions leave the grid[citation:1][citation:4].

4.1 The four actions from (1,1)

Action	Result	Reward	Next state
North	Off grid	-1	Stays at (1,1)
West	Off grid	-1	Stays at (1,1)
South	Moves to (2,1)	0	(2,1)
East	Moves to (1,2)	0	(1,2)

The values of the neighbors (from Figure 3.5):

- $v_\pi(2,1) = 1.5$ (the cell below)
- $v_\pi(1,2) = 8.8$ (the cell to the right)
- $v_\pi(1,1) = 3.3$ (the value of the corner cell itself, for the off-grid transitions)

4.2 Applying the Bellman equation

$$\begin{aligned} v_\pi(1,1) &= \frac{1}{4} \left[\underbrace{(-1 + 0.9 \cdot v_\pi(1,1))}_{\text{North: off grid}} + \underbrace{(-1 + 0.9 \cdot v_\pi(1,1))}_{\text{West: off grid}} \right. \\ &\quad \left. + \underbrace{(0 + 0.9 \cdot v_\pi(2,1))}_{\text{South}} + \underbrace{(0 + 0.9 \cdot v_\pi(1,2))}_{\text{East}} \right]. \end{aligned}$$

Substituting the known values $v_\pi(2,1) = 1.5$ and $v_\pi(1,2) = 8.8$:

$$\begin{aligned} v_\pi(1,1) &= \frac{1}{4} \left[(-1 + 0.9 \cdot 3.3) + (-1 + 0.9 \cdot 3.3) + (0.9 \cdot 1.5) + (0.9 \cdot 8.8) \right] \\ &= \frac{1}{4} \left[2(-1 + 2.97) + 1.35 + 7.92 \right] = \frac{1}{4} \left[2(1.97) + 9.27 \right] \\ &= \frac{1}{4} \left[3.94 + 9.27 \right] = \frac{13.21}{4} = 3.3025 \approx 3.3. \end{aligned}$$

This matches the value in Figure 3.5[citation:1][citation:4].

4.3 The key insight for corner cells

The critical point is that when an action would take the agent off the grid, the agent stays in the same state (so $s' = s$), and receives a reward of -1 [citation:8]. This is why the value $v_\pi(1, 1)$ appears on both sides of the Bellman equation for the off-grid actions. The equation is still a linear equation in $v_\pi(1, 1)$, and it can be solved as above.

5 The complete 5×5 value function

Solving the full system of 25 linear equations (one per cell) gives the value function shown in Figure 3.5 of Sutton & Barto:

	1	2	3	4	5
1	3.3	8.8	4.4	5.3	1.5
2	1.5	3.0	2.3	1.9	0.5
3	0.1	0.7	0.7	0.4	-0.4
4	-1.0	-0.4	-0.4	-0.6	-1.2
5	-1.9	-1.3	-1.2	-1.4	-2.0

This table matches the values printed in Figure 3.5 (right) and reproduced in various course materials[citation:3].

6 Understanding the special states A and B

The gridworld in Figure 3.5 has special states A and B that are not in the simple 5×5 grid positions described above. In the original S&B Figure 3.5, A and B are overlaid on specific cells:

- State A is at position (1,1) in the figure. From A, all four actions yield $+10$ and take the agent to A' (at position (1,5) in the figure).
- State B is at position (1,3). From B, all four actions yield $+5$ and take the agent to B' (at position (3,3))[citation:8].

However, in the 5×5 grid with values shown above (which is the one at page 60 that the Stack Overflow question refers to), the special states A and B are not included. That grid is a pure 5×5 grid with only the edge penalty (-1) and the equiprobable random policy.

The version with A and B is the one shown in Figure 3.5 (left) of S&B, which is a different grid layout (not the simple 5×5). The values shown in Figure 3.5 (right) are for that A/B version, and they include the effects of the $+10$ and $+5$ rewards from A and B.

The confusion arises because different sources reproduce different versions. The 5×5 grid with the values in the table above (3.3, 8.8, 4.4, 5.3, 1.5, ...) is the one without A and B, and it is the one discussed in the Stack Overflow question about corner cells[citation:1][citation:4].